

Heitor Marques Simões Barbosa

Ph.D. Student in Computer Science, Tulane University

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Education

Ph.D. Tulane University, Computer Science New Orleans, LA, USA
• Advisor: Dr. Hridesh Rajan, Laboratory for Software Design. Aug 2026 – present

B.S. University of Brasília (UnB), Software Engineering Brasília, Brazil
June 2020 – July 2025

Publications

DataFabric-DermAlert: A Tool for Federated Health Data Integration and Fairness-Aware Dataset Construction Sept 2026
Abdul Hannan, **Heitor Marques S. Barbosa**, Carla Silva Rocha Aguiar, Isaque Alves
(Brazilian Symposium on Software Engineering (SBES 2026), São Paulo, Brazil)

Research Experience

DermAlert Project, University of Brasília, Researcher Brasília, Brazil
Jan 2025 – July 2026
• Co-developed DataFabric-DermAlert, an open-source tool for federated health data integration, semantic equivalence management, and versioned, auditable dataset construction (SBES 2026).
• Contributed to a data fabric architecture for capturing, storing, and analyzing dermatology images in Brazil's Basic Health Units (UBS) to support early skin cancer screening.

PAULA Project, University of Brasília, AI Team Lead Brasília, Brazil
Mar 2023 – Dec 2023
• Led the AI team of PAULA (Paranoá Alfabetizando Usando Letramento Analógico), a UnB extension project that develops an offline mobile app, based on Paulo Freire's method, to help semi-literate and illiterate residents of Paranoá learn to read and write.
• Developed handwriting-recognition models (convolutional and transformer-based networks, stroke analysis) that read and correct letters written with the finger on the phone screen.
• Designed backend APIs to serve models and process data efficiently and securely.

AI.lab, University of Brasília, Machine Learning Researcher Brasília, Brazil
July 2022 – Dec 2023
• Worked on the artificial intelligence track of Justiça 4.0 (CNJ/UNDP), a program for the digital transformation of the Brazilian Judiciary, under the coordination of Dr. Nilton Correia da Silva and Dr. Fabrício Ataídes Braz.
• Implemented and evaluated clustering algorithms (K-means, GMM, spectral clustering, Kohonen networks, DBSCAN) to group similar court cases and discover topics for intelligent triage.
• Built text-preprocessing pipelines versioned with DVC and tracked experiments with MLflow; used multiprocessing to scale to large document collections.
• Collaborated with legal experts to validate labels, improving model accuracy and supporting the modernization of the judiciary.

Professional Experience

Forlex, AI Engineer

Remote

July 2025 – July 2026

- Designed and evaluated retrieval-augmented generation (RAG) pipelines for legal documents, benchmarking embedding models and chunking strategies to maximize context retrieval.
- Fine-tuned small language models (SLMs) for high-precision legal tasks, with a focus on data consistency and model performance.

SantoDigital, Machine Learning Specialist

Remote

Jan 2025 – July 2025

- Developed and trained computer vision and AI models for document processing (SantoID).
- Deployed models on Google Cloud Platform (Cloud Storage, Vertex AI, Cloud Run) and built API endpoints for integration with external systems.

Poder360, Data Engineer

Remote

Feb 2023 – Jan 2025

- Built a Celery-based automation system with scrapers that collect, process, and analyze large volumes of data from news agencies and government bodies in real time.
- Adopted MongoDB with indexing and query optimization, and deployed the system to production, enabling faster monitoring of emerging trends.

Brazilian Ministry of Education (MEC), Intern

Brasília, Brazil

Dec 2021 – June 2022

- Implemented computer vision models that recognize fields in historical medical residency record books and extract their handwritten content.
- Built business intelligence dashboards on medical and multidisciplinary residency programs, and designed an information security plan for sensitive data.

Volunteer

Debian Project, Contributor

Remote

Aug 2023 – Mar 2024

- Packaged and maintained software for the Debian stable and unstable branches in compliance with the Debian Policy.
- Automated build processes with CI/CD pipelines and debugged low-level compilation errors.

Technical Skills

Programming: Python, C/C++

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, spaCy, TRL, ms-swift

MLOps: MLflow, DVC, TensorBoard, model monitoring

Data and Cloud: PostgreSQL, MongoDB, MySQL, AWS, Google Cloud Platform

Software: FastAPI, Django, Docker, Kubernetes, CI/CD